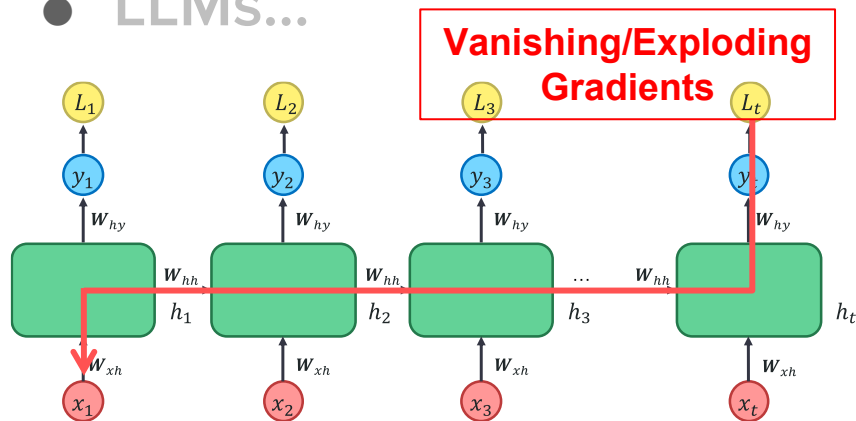


How to model complex dynamic systems in real time, with little data and low computational costs?

Yes, we are talking about time-series

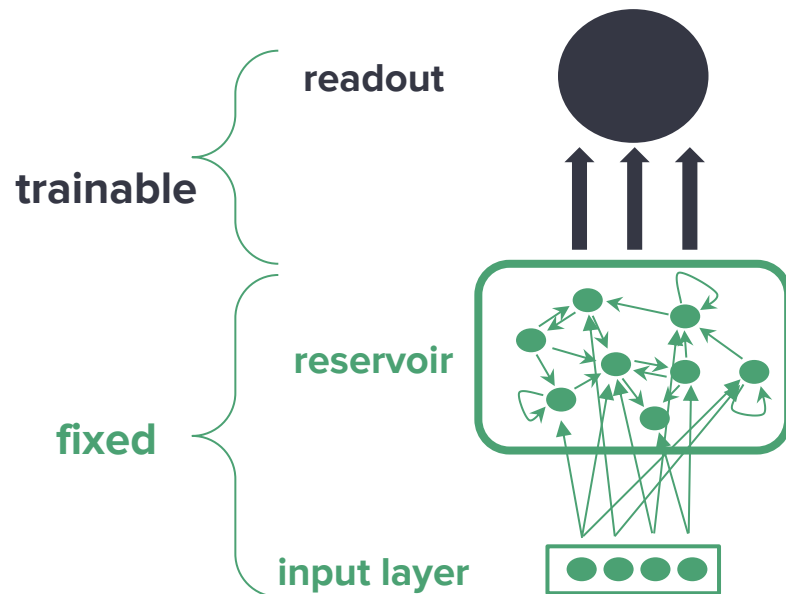
What people use

- RNNs
- Gated architectures (LSTM, GRU)
- Transformers
- LLMs...



What we use

- Echo State Networks



Lukoševičius, Mantas, and Herbert Jaeger. "Reservoir computing approaches to recurrent neural network training." *Computer science review* 3.3 (2009): 127-149.

The idea

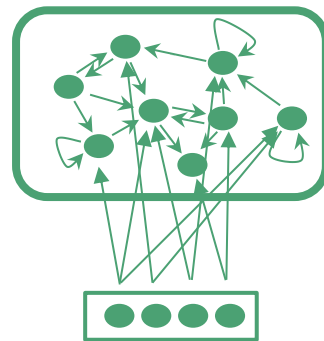
Key message: Separate dynamics and learning.

- **Reservoir:** a nonlinear dynamical system
It projects the input into a high-dimensional space
- **Training:** simple linear regression.
Closed-form / online (RLS)

Engineer the reservoir to be:

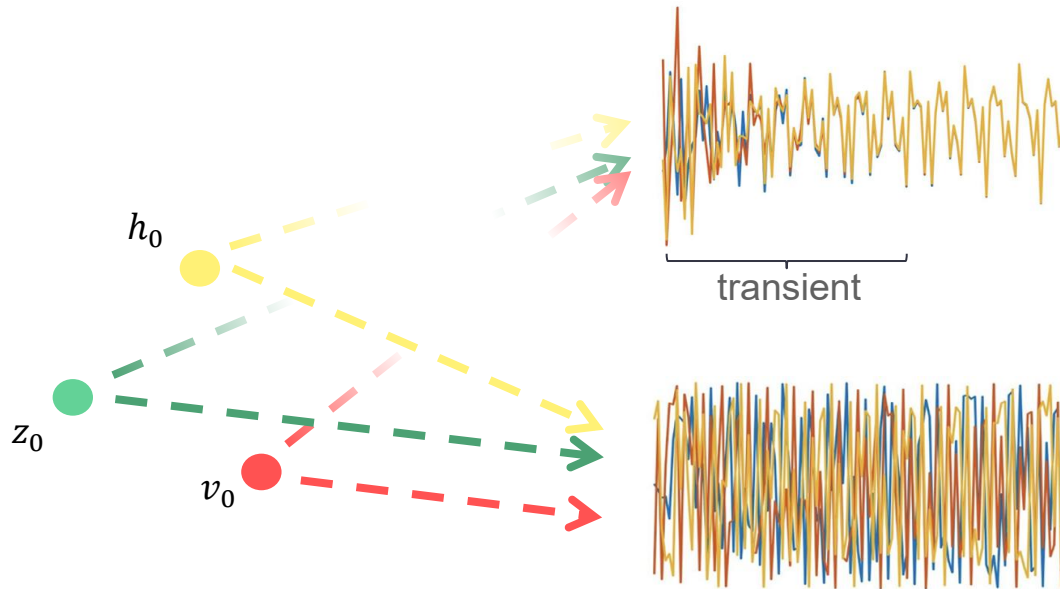
- Stable &
- Expressive

Trade-off



“Complex dynamics, simple learning.”

Stability in Practice



Stable dynamics

- orbits synchronize after a transient

Unstable dynamics

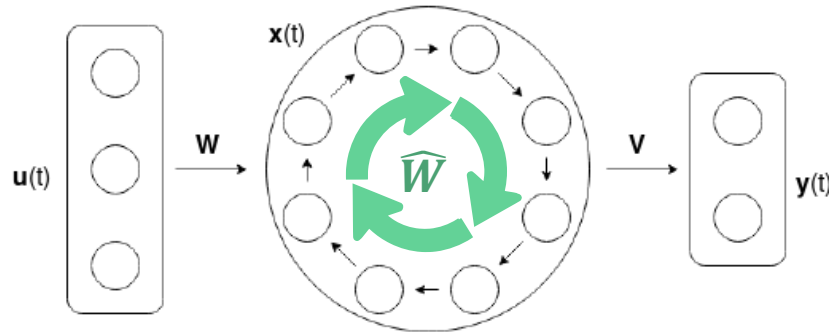
- orbits are sensitive to initial conditions

Expressivity

- **Entropy** of recurrent units activations
- Study the **short-term memory ability** of the system
- **Edge of stability/chaos**: reservoir at the border of stability
- ...

Schrauwen, B., Wardermann, M., Verstraeten, D., Steil, J.J. and Stroobandt, D., 2008. Improving reservoirs using intrinsic plasticity. *Neurocomputing*, 71(7-9), pp.1159-1171.

Ceni, Andrea, and Claudio Gallicchio. "Edge of stability echo state networks." *IEEE TNNLS* 2024.



$$\hat{W} = \begin{pmatrix} 0 & 0 & \dots & 0 & \hat{w} \\ \hat{w} & 0 & \dots & 0 & 0 \\ 0 & \ddots & \ddots & \vdots & \vdots \\ \vdots & \ddots & \ddots & 0 & 0 \\ 0 & \dots & 0 & \hat{w} & 0 \end{pmatrix}$$

Rodan, A. and Tino, P., 2010. Minimum complexity echo state network. *IEEE transactions on neural networks*, 22(1), pp.131-144.

ESNs are naturally suited to learning as data arrives

Advantages:

- training online
- Low cost computation
- More interpretability and control wrt deep RNNs

Some Use cases:

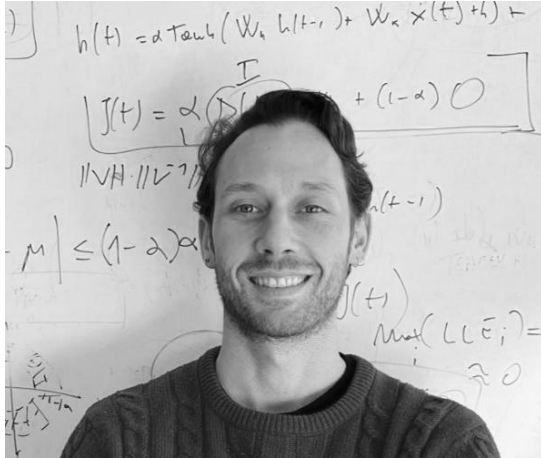
- Time-series classification
- Time-series forecasting
- Anomaly detection
- Edge computing

The Challenge

Can we exploit ESNs in the context of **Streaming and Continual Learning** scenarios?

Contact us

**Assistant Professor
at University of Pisa.**



Contact:

andrea.ceni@unipi.it

**Associate Professor
at University of Pisa.**



Contact:

claudio.gallicchio@unipi.it